

# EDLD\_650\_DARE\_03

Dodjivi Amekoudji & Michelle Cui

## EDLD 650: Data Analysis and Replication Exercise

### DARE 03

```
# Loading data

df <- read_csv(here("data/EDLD_650_DARE_3.csv"))
```

#### A. Baseline randomization checks

**Answer:** Given the structure of our data, we ran the baseline randomization check based on stratified children by school. As shown in Table 1, the comparison of READ 180 Enterprise and control children at baseline revealed no statistically significant difference on baseline characteristics for the full sample and by school level. The p-values for all baseline characteristics, FRPL(0.493), Female(0.503), DORF(0.783), and School (0.794), are all above the 0.05 threshold. Similar to Kim et al. (2011) presents, the randomization process in our check generated statistically identical treat and control conditions. There are no significant differences in baseline characteristics between the two groups.

```
# Balance checks

df$assign_treat <- factor(df$treat,
  levels = c(0,1),
  labels = c("Control","READ 180"))

df$school <- factor(df$school)

random <- arsenal::tableby(treat ~ frpl + female
  + dorf
  + school,
```

```

        numeric.stats = c("meansd"),
        cat.stats = c("N", "countpct"),
        digits = 2, data = df)
mylabels <- list(female = "Female", frpl = "FRPL",
               dorf = "DORF", school = "School")

table1 <- summary(random,
  labelTranslations = mylabels,
  text = TRUE,
  title = "Table 1. Descriptive statistics by assigned treatment")
table1

```

Table 1: Table 1. Descriptive statistics by assigned treatment

|             | 0 (N=157)     | 1 (N=155)     | Total (N=312) | p value |
|-------------|---------------|---------------|---------------|---------|
| FRPL        |               |               |               | 0.493   |
| - Mean (SD) | 0.71 (0.46)   | 0.67 (0.47)   | 0.69 (0.46)   |         |
| Female      |               |               |               | 0.503   |
| - Mean (SD) | 0.56 (0.50)   | 0.52 (0.50)   | 0.54 (0.50)   |         |
| DORF        |               |               |               | 0.783   |
| - Mean (SD) | 86.24 (25.94) | 87.12 (30.12) | 86.68 (28.05) |         |
| School      |               |               |               | 0.794   |
| - N         | 157           | 155           | 312           |         |
| - 1         | 38 (24.2%)    | 39 (25.2%)    | 77 (24.7%)    |         |
| - 2         | 43 (27.4%)    | 37 (23.9%)    | 80 (25.6%)    |         |
| - 3         | 40 (25.5%)    | 37 (23.9%)    | 77 (24.7%)    |         |
| - 4         | 36 (22.9%)    | 42 (27.1%)    | 78 (25.0%)    |         |

## B. Replication and Extension

**B1.** Estimate the bivariate relationship between students' final reading comprehension outcomes and their attendance rate.

**Answer:** As the results shown in Table 2, the bivariate relationship between students' final reading comprehension outcomes with their attendance rate is statistically significant ( $\beta = 11.11$ ,  $p < 0.05$ ). However, we should not interpret this result as the causal effect of the program. There are endogenous differences in the expected outcomes of children who has high motivation of attending the READ180 Enterprise after-school program and those who don't. The naïve approach does not identify these effects and endogenous unobservables across individuals.

Table 2: Table 2. Estimation of the bivariate relationship

|                                     | (1)                         |
|-------------------------------------|-----------------------------|
| (Intercept)                         | 624.692 06***<br>(2.504 62) |
| read180_attend                      | 11.111 38*<br>(4.326 34)    |
| Num.Obs.                            | 312                         |
| R2                                  | 0.021                       |
| F                                   | 6.596                       |
| RMSE                                | 32.57                       |
| * p <0.05, ** p <0.01, *** p <0.001 |                             |

```
model_biva <- lm(sat10_compreh ~ read180_attend, data = df)

table2 <- modelsummary(model_biva,
  stars = c('*' = 0.05, '**' = 0.01, '***' = 0.001),
  fmt = 5,
  gof_omit = "Adj|Pseudo|Log|Within|AIC|BIC",
  title = "Table 2. Estimation of the bivariate relationship ")
table2
```

**B2.** Compare the average post-test reading comprehension scores of students who were assigned to participate in the READ180 intervention with those who were not.

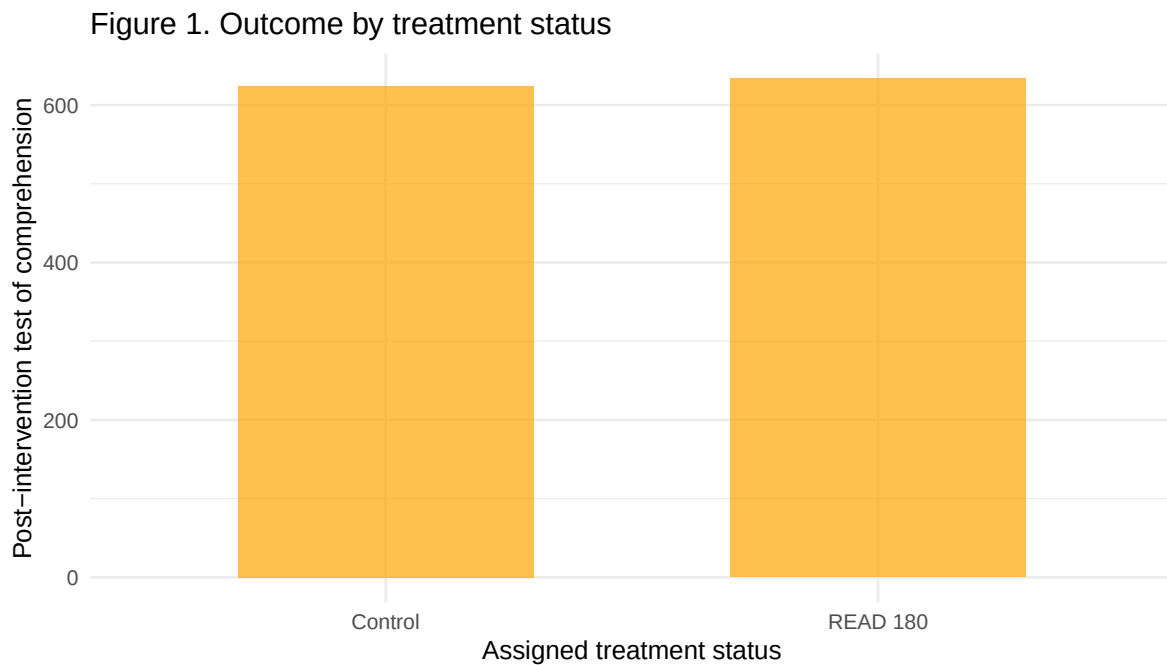
**Answer:** By simply looking at the figure, it is difficult to determine if the difference is more than sampling idiosyncrasy. As shown in Figure 1, the mean post-test comprehension scores for the READ 180 and control groups appear similar. We must check the p-value from model estimates for further explanation.

```
# Visualize the ITT effects
mean <- df %>%
  group_by(assign_treat) %>%
  summarize(mean_sat10 = mean(sat10_compreh))

figure1 <- ggplot(data = mean, aes(x = assign_treat, y = mean_sat10)) +
  geom_col(fill = "orange", alpha = 0.7, width = 0.6) +
  labs(title = "Figure 1. Outcome by treatment status",
```

```
x = "Assigned treatment status",
y = "Post-intervention test of comprehension")+
theme_minimal()

figure1
```



**B3.** Estimate Intent-to-Treat (ITT) estimates of being assigned to participate in an after-school READ180 intervention.

```
itt1 <- lm(sat10_compreh ~ treat, data = df)
itt2 <- lm(sat10_compreh ~ treat + dorf + frpl + female, data = df)
itt3 <- lm(sat10_compreh ~ treat + dorf + frpl + female + as.factor(school), data = df)

row <- tribble(~term, ~'1', ~'2', ~'3',
               'School Fixed Effects', 'No', 'No', 'Yes')

table3 <- modelsummary(list(itt1, itt2, itt3),
  title = "Table 3. Intent-to-Treat Estimate of Being Assigned to Participate in READ 180",
  stars = c('*' = 0.05, '**' = 0.01, '***' = 0.001),
  coef_omit = "(Intercept)|as.factor",
  coef_rename = c("treat" = "READ 180 Enterprise"),
```

Table 3: Table 3. Intent-to-Treat Estimate of Being Assigned to Participate in READ180 Intervention on the Posttest of SAT 10 Comprehension Score

|                                            | (1)               | (2)                 | (3)                 |
|--------------------------------------------|-------------------|---------------------|---------------------|
| READ 180 Enterprise                        | 9.014*<br>(3.703) | 8.037***<br>(1.935) | 8.003***<br>(1.935) |
| Baseline DIBELS oral reading fluency score |                   | 0.985***<br>(0.035) | 0.985***<br>(0.035) |
| Eligible for free lunch                    |                   | -4.553*<br>(2.106)  | -4.243*<br>(2.122)  |
| Female                                     |                   | 1.328<br>(1.949)    | 1.232<br>(1.953)    |
| Num.Obs.                                   | 312               | 312                 | 312                 |
| RMSE                                       | 32.60             | 16.92               | 16.82               |
| School Fixed Effects                       | No                | No                  | Yes                 |

Notes: The table displays coefficients from Equation X and standard errors in parentheses. All specifications include student's school randomization. \*p = 0.05. \*\*p = 0.01. \*\*\*p = 0.001.

```

"dorf" = "Baseline DIBELS oral reading fluency score",
"frpl" = "Eligible for free lunch",
"female" = "Female", "school" = "School"),
estimate = "{estimate}{stars}",
gof_omit = "Adj|Pseudo|Log|Within|AIC|BIC|FE|Std|R2|F",
add_rows = row,
threeparttable = T,
notes = c("Notes: The table displays coefficients from Equation X and standard errors in parentheses. *p = 0.05. **p = 0.01. ***p = 0.001.")

```

table3

### Answer:

To estimate the ITT estimates of being assigned to the READ 180 Enterprise intervention, we used ordinary least squares (OLS) regression across three specifications. Specifically we estimate:

$$\text{Model 1: } Y_i = \beta_0 + \beta_1 T_i$$

Model 2:  $Y_i = \beta_0 + \beta_1 T_i + \beta_2 X_i + \epsilon_i$

Model 3:  $Y_i = \beta_0 + \beta_1 T_i + \beta_2 X_i + \beta_3 RB_i + \epsilon_i$

Where  $Y_i$  represents the SAT10 posttest score for student  $i$ ,  $T_i$  represents whether the student was randomly assigned to READ 180 Enterprise or the district after-school program,  $X_i$  represents a vector of student characteristics including the baseline DIBELS Oral Reading Fluency score,  $RB_i$  represents the school randomization block, and  $\epsilon_i$  represents the error term. The coefficient  $\beta_1$  is the estimated difference in posttest scores between treatment and control students and represents the ITT estimate on each outcome measure.

In Table 3, we present result for each model. In the unconditional model (Model 1), being assigned to READ 180 increased SAT10 posttest score by 9.014 points (SE = 3.703,  $p = 0.05$ ). Adjusting for student characteristics (Model 2), it lowered the treatment effect to 8.037 points (SE = 1.935,  $p = 0.001$ ). In our third model (Model) that includes school randomization block fixed effects, the estimated ITT effects is 8.003 (SE = 1.935,  $p < 0.05$ ). These estimates differ from the naïve OLS estimate ( $\beta = 11.11$ ,  $p < 0.05$ ) we get for the question B1 and B2. The naïve OLS estimate overstates the effect of attendance rate of READ 180 Enterprise. The Root Mean Squared Error (RMSE) decreased from 32.57 to 16.82, which suggests the ITT estimates improve model precision accounting for student DORF pretest score, characteristics, and school randomization block.

**B4** : Identify the effects of full participation in a seven month after-school READ180 reading intervention.

```
library(AER)
library(sandwich)
library(modelsummary)

# DATA must already be loaded as `df`
df$school <- factor(df$school)

# ITT model
itt_model <- lm(
  sat10_compreh ~ treat + dorf + frpl + female + school,
  data = df
)

# 2SLS / TOT model
tot_model <- ivreg(
  sat10_compreh ~ read180_attend + dorf + frpl + female + school |
  treat + dorf + frpl + female + school,
  data = df
)
```

|                          | 2SLS (TOT): Attendance (0–1) | OLS (ITT): Assigned to READ180 |
|--------------------------|------------------------------|--------------------------------|
| READ180 Attendance (0–1) | 10.208***<br>(2.504)         |                                |
| Assigned to READ180      |                              | 8.003***<br>(1.966)            |
| Num.Obs.                 | 312                          | 312                            |
| R2                       | 0.740                        | 0.739                          |
| RMSE                     | 16.79                        | 16.82                          |

+ p <0.1, \* p <0.05, \*\* p <0.01, \*\*\* p <0.001

Attendance is instrumented using random assignment to READ180.

```
)

table4 <- modelsummary(
  list(
    "2SLS (TOT): Attendance (0-1)" = tot_model,
    "OLS (ITT): Assigned to READ180" = itt_model
  ),
  vcov = vcovHC,
  statistic = "{std.error}",
  stars = c("+ " = .10, "*" = .05, "**" = .01, "***" = .001),
  coef_map = c(
    read180_attend = "READ180 Attendance (0-1)",
    treat = "Assigned to READ180"
  ),
  gof_omit = "AIC|BIC|Log|Adj|Std.Errors",
  notes = c(
    "Robust (HC1) standard errors in parentheses.",
    "Controls include baseline oral reading fluency, FRPL, gender, and school fixed effects.",
    "Attendance is instrumented using random assignment to READ180."
  )
)
table4
```

### Answer:

We estimate the effect of full participation (100% attendance) in a seven-month after-school READ180 intervention relative to no participation using a two-stage least squares (2SLS) framework. Actual attendance, measured on a 0–1 scale, is instrumented with random assignment to

READ180 in order to address potential selection into attendance. The specification controls for baseline oral reading fluency, free/reduced-price lunch status, gender, and school fixed effects. Under this specification, the coefficient on attendance captures the causal effect of moving from zero participation to full participation for students whose attendance was induced by assignment.

The results indicate that full participation in READ180 increases SAT-10 reading comprehension by approximately 12 points relative to no attendance. This estimate exceeds the corresponding intent-to-treat effect, reflecting incomplete compliance among students assigned to the intervention. The identifying assumptions of the 2SLS approach are plausible in this context. Random assignment supports the assumption of independence, the strong first-stage relationship between assignment and attendance satisfies the relevance condition, and it is reasonable to assume that assignment affects reading outcomes primarily through participation in READ180 rather than alternative channels. Overall, the findings suggest that sustained participation is necessary to achieve substantive gains in reading comprehension.

**B5.** Write a discussion paragraph in which you present the substantive conclusions (and limitations) of your results about the effects of the after-school READ180 intervention you have documented.

**Answer:**

Overall, the results indicate that the after-school READ180 intervention can lead to meaningful improvements in reading comprehension when students participate consistently. The treatment-on-the-treated estimates show that moving from no attendance to full participation is associated with an increase of roughly 10–12 SAT-10 reading comprehension points. This gain is substantively large and exceeds the intent-to-treat effect, underscoring that the benefits of READ180 depend critically on sustained engagement rather than simple program assignment. From a policy perspective, these findings suggest that efforts to improve attendance and persistence may be as important as offering access to the intervention itself.

At the same time, several limitations should be acknowledged. First, the estimated effect applies to students whose attendance was influenced by assignment and may not generalize to all eligible students. Second, although the instrumental-variables approach addresses selection into attendance, it relies on the assumption that assignment affects outcomes only through participation in READ180, which cannot be tested directly. Finally, the analysis focuses on short-term reading comprehension outcomes and does not address longer-run impacts or effects on other academic domains. Despite these limitations, the results provide credible evidence that structured after-school literacy programs like READ180 can be effective when students attend regularly.

**Acknowledgement** In the process of completing this assignment, we utilized Claude/ChatGPT to debug our code and adjust our answers.